

Engineering Design Document: System Mathematics & Analog Conditioning

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This document details the theoretical calculations, physical principles, and integrated design constraints governing the analog signal conditioning pipeline and digital logic bridge.

1. High-Pass Filter & Virtual Ground Stage

Raw acoustic signals from the electret microphone arrive as Alternating Current (AC) waves superimposed on a 1.48V DC baseline. To amplify this wave without clipping against the ground rail, the operational amplifier requires a synthetic midpoint.

A voltage divider constructed from two 10 kΩ resistors biases the 3.3V supply to establish a stable 1.65V virtual ground. This specific resistance value provides a sufficiently stiff bias current to the op-amp input while minimizing parasitic power loss from the main supply.

A 0.1 μF coupling capacitor acts as a DC-blocking filter between the transducer and the pre-amp. To steady 0 Hz DC signals, the capacitor presents infinite resistance, completely isolating the microphone's native bias. Conversely, it presents near-zero resistance to high-frequency AC audio transients. The equivalent resistance of the voltage divider is calculated as:

$$R_{eq} = \frac{10000 \cdot 10000}{10000 + 10000} = 5000 \Omega$$

This yields a lower cut-off frequency of:

$$f_c = \frac{1}{2\pi \cdot R_{eq} \cdot C} = \frac{1}{2\pi \cdot 5000 \cdot (0.1 \times 10^{-6})} \approx 318.3 \text{ Hz}$$

This 318.3 Hz cut-off intentionally attenuates low-frequency environmental noise, such as HVAC rumble or footsteps, while allowing high-frequency acoustic transients—such as a hand clap—to pass into the amplification stage unimpeded.

2. Pre-Amplifier Gain Stage (Non-Inverting)

The filtered acoustic transient requires voltage scaling to be readable by logic gates. This is achieved by configuring the Op Amp as a non-inverting amplifier, where the gain (A_v) is governed by the feedback resistor (R_f) and the input resistor (R_{in}):

$$A_v = 1 + \frac{R_f}{R_{in}}$$

The hardware utilizes a 10 k Ω potentiometer for R_f and a 100 Ω fixed resistor for R_{in} , which is placed in series with a 10 μ F capacitor to ground.

At the 0 Hz DC baseline, the 10 μ F capacitor acts as an open circuit, driving the denominator toward infinity and resulting in a unity gain (1 $\times$).

This ensures the 1.65V virtual ground is perfectly stable and not erroneously amplified into the voltage rails. For fast-moving AC transients exceeding 1000 Hz, the capacitor acts as a short circuit, allowing the system to achieve its maximum operational gain:

$$A_v = 1 + \frac{10000}{100} = 101 \times \text{Gain}$$

This 101x multiplier successfully scales a nominal 20 mV acoustic spike into a 2.2V AC swing, preparing it for the logic thresholding stages.



3. Envelope Detector & Unity Gain Buffer

Because digital microcontrollers cannot efficiently process jagged, bidirectional AC waves, this stage rectifies and stretches the transient into a sustained DC logic level. A half-wave rectifier diode blocks reverse current flow, stripping the negative half of the AC wave and passing only the positive voltage peaks.

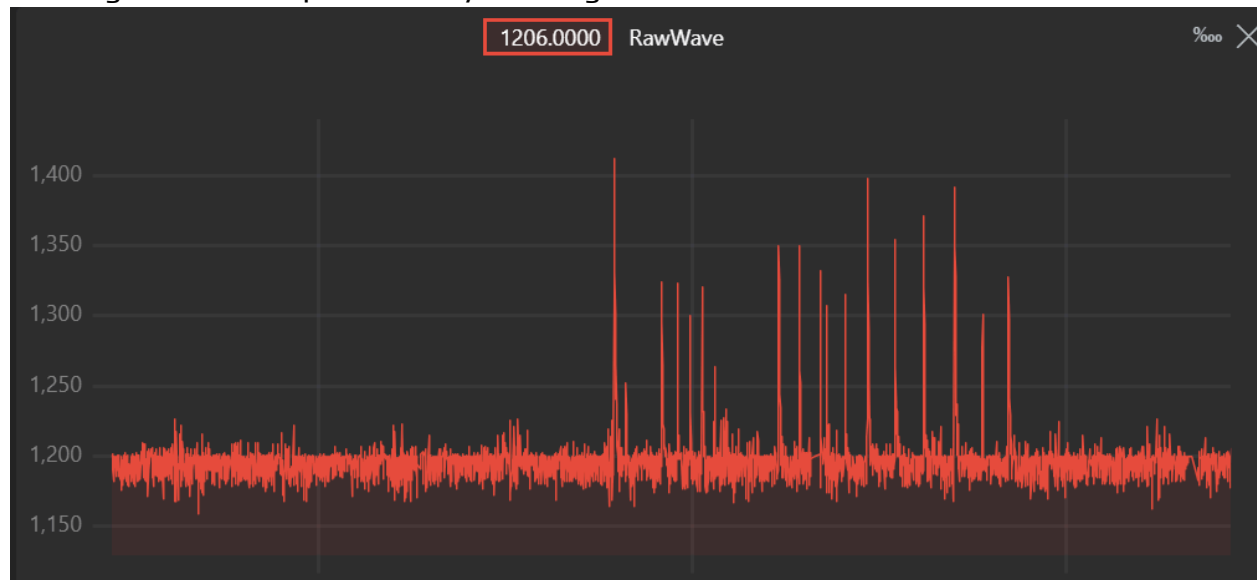
These positive peaks charge a parallel RC tank circuit. Due to the diode blocking reverse flow, the 10 μ F capacitor must discharge exclusively through a 10 k Ω parallel resistor to ground. The discharge speed is dictated by the time constant (τ):

$$\tau = R \cdot C = 10000 \cdot (10 \times 10^{-6}) = 0.1 \text{ seconds}$$

An acoustic clap lasts less than 5 milliseconds, meaning a raw peak could easily be missed by a microcontroller interrupt.

The 0.1-second time constant physically stretches the microsecond event into a wide, sustained DC envelope, guaranteeing digital capture.

The envelope then passes through a unity gain buffer wired as a voltage follower. This provides infinite input impedance and zero output impedance, acting as a robust power supply for the signal. It strictly prevents the subsequent logic stages from drawing current and prematurely draining the RC tank.



4. Schmitt Trigger Digitization

To transition from analog conditioning to digital logic without introducing signal chattering, the final op-amp is configured as a Schmitt trigger. The integrated resistor network establishes dual trip points (hysteresis) dictated by the feedback equation:

$$V_{+} = V_{buffer} + \left[(V_o - V_{buffer}) \times \left(\frac{R_{in}}{R_{in} + R_f} \right) \right]$$

This configuration establishes a strict upper trip point of 1.05V and a lower trip point of 0.99V.

By requiring the rising envelope to cross 1.05V to trigger a HIGH state, and forcing it to decay below 0.99V before returning to a LOW state, the hardware enforces a singular, bounce-free digital edge.

This dual-threshold approach renders the system entirely immune to room echoes or minor voltage fluctuations hovering near the baseline.

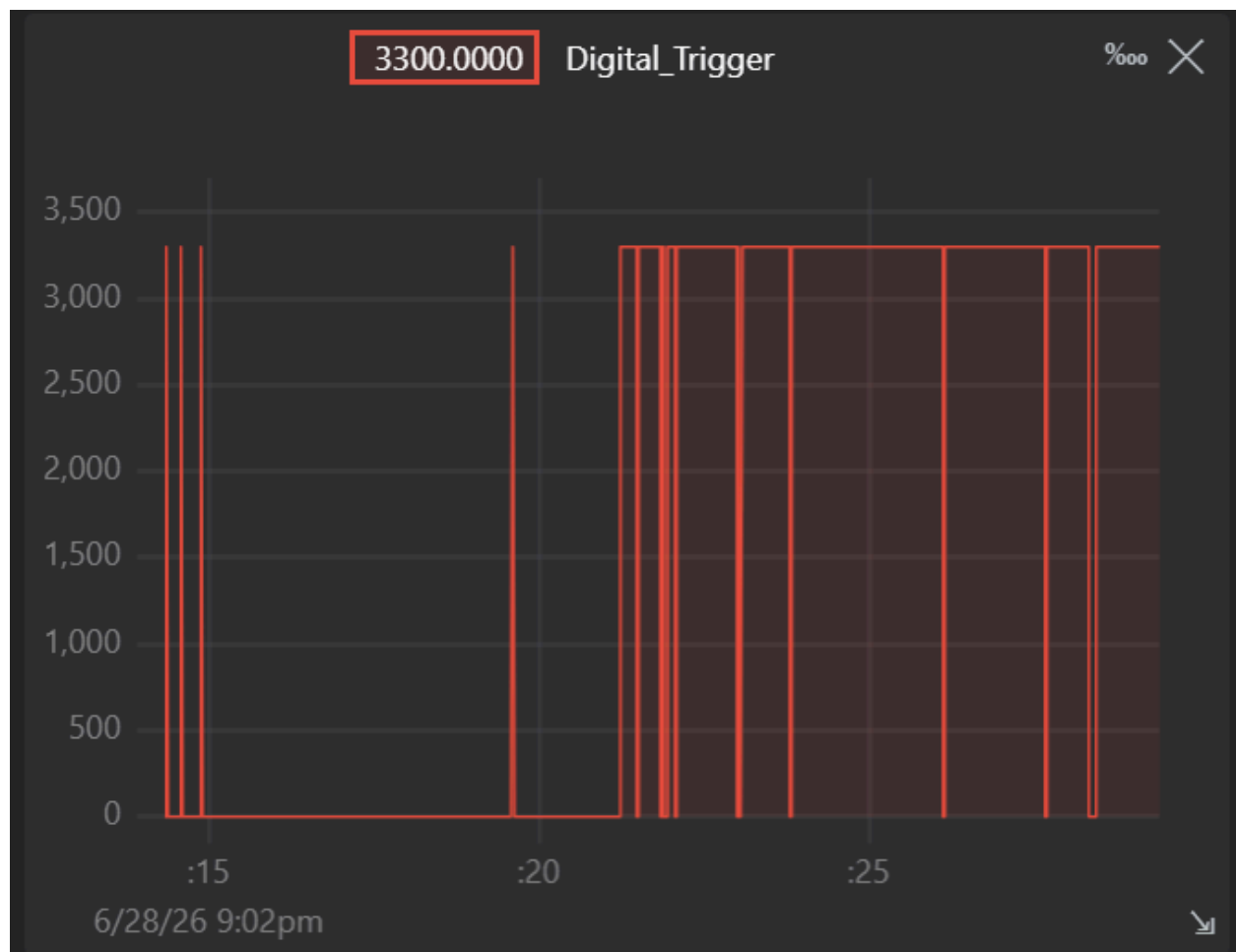
5. Logic Level Bridge & Isolation Constraints

The Schmitt trigger generates a 1.95V digital output edge. This voltage level is unstable and unreliable for driving standard 3.3V logic processors.

A discrete 2N2222 NPN transistor is deployed as an isolating logic level shifter to safely bridge this gap.

The system enforces a strict hardware design constraint at this bridge: the collector of the NPN transistor is tied to the 3.3V logic domain exclusively via a physical 5.1 k Ω external pull-up resistor.

Internal microcontroller pull-ups are strictly prohibited in this design architecture. The physical 5.1 k Ω pull-up guarantees a pristine, high-current logic HIGH when the transistor is in cutoff, completely eliminating floating inputs and ensuring industrial-grade signal stability before the hardware interrupt is fired.



6. Solid-State Actuation & Inductive Load Protection

The pristine digital edge captured by the logic stage triggers the ESP32 microcontroller, which is responsible for driving an N-Channel MOSFET (M1) to control a 5V Brushless DC (BLDC) fan. The microcontroller interfaces with the MOSFET gate via a 330 Ω series resistor. This specific resistance limits the instantaneous inrush current required to charge the MOSFET's parasitic gate capacitance, protecting the logic pins from overcurrent transients while ensuring rapid switching.

Because the BLDC fan represents a highly inductive load, the sudden interruption of current by the MOSFET going into cutoff will generate a destructive high-voltage

back-electromotive force (back-EMF) spike. To mitigate this, a 1N4004 flyback diode is placed in parallel across the fan load. When the MOSFET disengages, this diode provides an immediate, safe recirculation path for the inductive kickback, clamping the transient voltage and preventing avalanche breakdown of the MOSFET.

7. Visual Status Indication

To provide immediate, hardware-level diagnostic feedback without relying on serial monitors, the system features a dedicated status bar. This consists of Yellow and Green LEDs, each driven by isolated logic pins. To ensure the forward current remains well within the safe operational limits of both the semiconductor diodes and the logic controller, each LED is placed in series with a $220\ \Omega$ current-limiting resistor tied directly to ground.